

The Role of Longevity Medicine in Reforming Primary Healthcare

Abstract

Publicly financed health systems face an urgent convergence of fiscal, demographic, and epidemiological pressures, including population ageing, multimorbidity, lifestyle-related chronic disease, and rising labor-intensive costs, that render predominantly reactive care models economically unsustainable. This chapter argues that longevity medicine—anchored in primary prevention, predictive risk stratification, validated biomarkers, and multidisciplinary, data-enabled care—offers a credible strategy to extend healthy life years, reduce chronic disease burden, and optimize resource allocation. We emphasize practical implementation through publicly accessible longevity clinics, operationalized with standardized assessment, stepped-care pathways, integrated digital records, and measurable functional and economic endpoints. Recognizing the limits of monolithic macro-reform, we propose a hybrid policy mix organized via the NATO framework (Nodality–Authority–Treasure–Organization), sequenced to balance early wins, fiscal realism, and workforce readiness: information infrastructures and citizen-facing tools; regulatory and accreditation standards for biomarkers and clinics; prevention-focused financing and equitable subsidies; organizational delivery models through community hubs and carefully governed PPPs; and workforce development to support multidisciplinary, data-literate teams. Additional considerations include phased piloting, pragmatic evaluation using healthspan-relevant outcomes, and climate-adaptive interventions to protect vulnerable populations, highlighting a systemic, evidence-driven roadmap for sustainable public health reform in the era of longevity.

Keywords: Longevity medicine · Healthspan; Biomarkers of ageing · Risk stratification · Longevity clinics · Prevention; Primary care · Hybrid policy · Polycentric governance · Public–private partnerships · Longevity policy

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Economic Unsustainability of the Public Health System

Health spending has outpaced economic growth for much of the past twenty years. The drivers are familiar: faster medical-price inflation, a steady widening of therapeutic options, and the rising weight of chronic disease. Across the 38 OECD countries, per-capita outlays now average just under USD 5,000 and exceed USD 12,500 in the United States—even after adjusting for purchasing-power differences (OECD 2023). Over the same period, governments have accumulated unusually high debt, and real interest rates have climbed at the fastest clip in a generation; together, these trends leave far less fiscal room to absorb further cost pressures (IMF 2024).

Population ageing and the changing epidemiology of disease together produce a fiscal shock that is at once large, persistent, and multi-dimensional. The absolute number of older adults is growing rapidly worldwide and, even where life expectancy has recovered from short-term calamities, healthy life expectancy (HALE) often lags overall gains — implying an expanded period of life lived with chronic morbidity and disability (WHO 2015; Global Burden of Disease Collaborative Network [GBD] 2019). In parallel, medical-price inflation, the rapid diffusion of new technologies and therapies, and rising wages in health professions have pushed public health spending upward even in periods of muted GDP growth (OECD 2023; IMF 2024). These forces combine to shrink the fiscal space available to governments just when the demand for long-term and chronic care is expanding.

The demographic turn toward older populations is already pushing up demand for health services—and with it, public outlays. Systems with a larger share of people aged 65 and over tend to spend markedly more per capita on healthcare (OECD 2021). At the same time, the dependency ratio rises: fewer workers are supporting more retirees, which concentrates the tax burden on a shrinking base. It is no surprise, then, that the World Health Organization cautions against business-as-usual; without more sustainable long-term care arrangements and credible preventive policies, fiscal pressures risk becoming unmanageable (WHO 2015).

From an expenditure standpoint, the concentration of cost among people with multiple chronic conditions is especially consequential. Across high-income and many middle-income settings, a small share of the population—people with multimorbidity and frailty—accounts for a disproportionate share of resource use: hospital admissions, long-term care, and complex outpatient management. This concentration creates both opportunity and risk: targeted upstream prevention and systematic case-finding can deliver large savings if evidence-based interventions delay or compress morbidity, but failing to act locks systems into costly reactive pathways that increase per-capita outlays and reduce equity (CDC 2022; GBD 2019).

Layered onto ageing is the surge in lifestyle-related chronic disease—cardiovascular conditions, diabetes, obesity, and a growing set of neurological disorders. These illnesses are now more prevalent than infectious disease and, crucially, far costlier to manage over time. In the United States, the Centers for Disease Control and Prevention estimates that about 90% of the nation's USD 4.1 trillion in annual health expenditures is accounted for by people living with chronic and mental health conditions (CDC 2022). In universal systems, much of that bill lands with the public sector. Yet many public models still default to reactive care, underinvesting in prevention and early intervention; the result is a slow but steady accumulation of structural inefficiencies (WHO 2014).

Healthcare workforce dynamics add a further layer of complexity. Effective public systems depend on sufficient numbers of skilled, fairly compensated professionals, but many countries report persistent shortages in nursing, primary care and rehabilitation, and caregiving more broadly. These gaps translate into heavier caseloads, higher turnover, burnout, and uneven quality — all of which reduce productivity and increase unit costs (Buchan, Duffield, & Jordan 2015). To recruit and retain staff—especially in shortage specialties and underserved regions—systems must raise salaries and enhance benefits, which in turn pushes up overall spending. This pattern reflects the classic economic mechanism known as Baumol’s cost disease, where labour-intensive sectors (like healthcare) experience rising costs because productivity gains lag behind wages in the broader economy (Baumol & Bowen 1966). In practice, Baumol’s argument helps explain why labour becomes the dominant component of health expenditure in ageing societies and why fiscal pressures persist even in the absence of obvious inefficiency.

Finally, three health-system inefficiencies amplify fiscal strain and compound the demographic and epidemiological drivers already described. First, workforce shortages and burnout—particularly in nursing, primary care, and rehabilitation—increase unit costs and reduce system productivity (Buchan et al. 2015). Second, clinical inefficiencies such as defensive medicine and overdiagnosis produce low-value care and unnecessary investigations (Mello et al. 2010; Welch et al. 2011). Defensive medicine—often motivated by legal risk or a risk-averse culture—leads clinicians to order tests or procedures with marginal clinical value, inflating costs without commensurate health gains (Mello et al. 2010). Overdiagnosis, frequently propelled by poorly targeted screening programs, results in treatment of conditions that would never have caused clinical harm, adding to patient burden and system waste (Welch et al. 2011). Third, moral-hazard overutilization in publicly funded delivery models—where care is provided at little or no cost to users—can weaken individual incentives for preventive behaviour and raise demand for low-value services (Zweifel & Manning 2000). Together these dynamics distort care priorities, create congestion in services, and further undermine the economic sustainability of public healthcare (Thomson et al. 2017).

Taken together, the demographic shift, the concentration of multimorbidity, rising medical prices and wages, and the three systemic inefficiencies narrow the policy window for sustainably “bending the cost curve.” Rather than relying solely on blunt price controls or program cuts, the evidence suggests policymakers must prioritize prevention, service redesign, workforce planning, and the careful adoption of technologies that demonstrably improve productivity and healthspan (OECD 2023; IMF 2024; WHO 2014).

2) Reframing Public Health Through Longevity Medicine

In light of the growing economic unsustainability of public health systems, longevity medicine emerges as an innovative model grounded in primary prevention, personalized medicine, and proactive clinical risk management (Zhavoronkov 2019). By integrating predictive analytics with targeted interventions, this approach (Sagner et al. 2017) directly addresses the structural drivers of the aforementioned unsustainability: it mitigates the impact of population ageing by preserving functional autonomy (Olshansky et al. 2007), lowers the prevalence of chronic diseases through targeted preventive strategies (World Health Organization [WHO] 2014), and curbs reliance on costly technologies by acting early on risk (Topol 2019). Moreover, it optimizes the allocation of health resources by concentrating on high-value interventions (Porter 2010) and enables more efficient use of clinical staff (Yu-Xuan Lyu 2024). Adopting

scientific, biomarker-based protocols can also curb defensive medicine and overdiagnosis by anchoring decisions in objective data (Alaoui-Jamali & Bijian 2018). In parallel, equipping citizens with education and continuous monitoring fosters agency and shared responsibility, mitigating free-riding and nurturing a more informed health culture (Kickbusch et al. 2016).

Longevity medicine reframes public health goals around the explicit aim of extending healthspan - the years lived in functional autonomy—rather than merely extending survival. To move from concept to practice, three interlocking capabilities must be operationalized.

First, validated biomarkers and stratification tools are needed to identify biological aging trajectories and to select individuals and populations most likely to benefit from interventions. Composite measures such as epigenetic clocks, proteomic signatures, and integrative physiological indices are maturing rapidly, but policy frameworks must set rigorous evidence thresholds—covering analytic validity, clinical validity, and clinical utility—before routine clinical adoption (Levine et al. 2018; Moqri et al. 2023).

Second, service models must translate biomarker and phenotypic data into actionable prevention. Publicly funded longevity clinics embedded within primary-care networks represent a pragmatic model: they could deliver midlife risk-stratified assessments, personalized prevention plans (including nutrition strategies, targeted medications where evidence supports their use, exercise and rehabilitation prescriptions), medication reviews, and ongoing remote monitoring. Crucially, these services should be evaluated not simply by single-disease endpoints but by functional outcomes such as intrinsic capacity, frailty indices, and major adverse events, as well as by cost-effectiveness (Bonnes et al. 2024; Topol 2019).

Third, evidence generation itself must adapt. Because definitive lifespan trials are neither feasible nor practical at population scale, pragmatic adaptive trials using validated healthspan surrogates—such as immune resilience, composite functional decline, or epigenetic age change—offer a credible path forward. Governments and funders, particularly through international geroscience consortia and organizations such as the U.S. National Institute on Aging (NIA), should prioritize midlife and older-adult trials incorporating multi-omics, environmental exposure data, and standardized functional outcomes to produce comparable and scalable insights.

Against this backdrop, longevity-oriented prevention clinics offer a practical vehicle for scaling the longevity-medicine paradigm. Integrating these facilities into public systems is essential to translate innovative prevention into standardized, validated services accessible beyond private or elite settings. As Bonnes et al. (2024) emphasize, such clinics operate through multidisciplinary teams that combine preventive medicine with advanced diagnostics, molecular biomarkers, genomics, epigenetics, and personalized digital assessments for risk stratification. This model supports the early detection of functional and physiological decline and enables targeted, evidence-based strategies to forestall the onset or progression of age-related conditions—including cardiovascular, oncological, metabolic, and neurodegenerative diseases (Bonnes et al. 2024; Zhavoronkov 2019; Sagner et al. 2017).

Public longevity clinics, operating in parallel with conventional reactive care, can act as a system-level lever to curb costs over the medium to long term. By identifying risk trajectories before they crystallize into overt disease, these clinics enable earlier—and typically less costly—interventions, relieving pressure on hospitals, long-term care, and high-priced

therapies (Olshansky et al. 2007; WHO 2014). In addition, the structured use of predictive algorithms, longitudinal monitoring, and digital tools supports a sustainable form of precision medicine rooted in objective data, helping to limit unwarranted clinical variation, overdiagnosis, and overtreatment (Topol 2019). These facilities thus emerge as pivotal nodes within a proactive health system capable of integrating medicine, data science, and individual behavior into a new architecture of public health (Shigekawa et al. 2018). Such reform may represent a turning point for ensuring the future sustainability of health systems, combining innovation, prevention, and universalism in a vision aligned with the demands of the ongoing demographic and technological transition (Kickbusch et al. 2016).

The hypothesis of a public, longevity-oriented reform does not arise in a theoretical or experimental vacuum; it rests on a growing international literature and on initiatives already launched in advanced healthcare contexts. The definition of clinical criteria, diagnostic protocols, and operational models for longevity clinics is the subject of recent studies aimed at their standardization and integration into public health systems. In particular, Maier (2024) proposes a conceptual and operational framework for developing healthcare structures dedicated to the promotion of longevity, outlining minimum criteria for clinical quality, scientific validation of biomarkers, and outcome evaluation.

These suggested frameworks are not based on abstractions but on real-world cases demonstrating feasibility and effectiveness in the public sector. In Israel, the Sheba Longevity Center, housed within the Sheba Medical Center—one of the world's most advanced hospitals—stands as an example of integration among personalized medicine, prevention, and clinical innovation in a public setting. Similarly, in Singapore, Alexandra Hospital has implemented a systemic approach to longevity medicine with the aim of extending the population's healthy life expectancy through multidisciplinary programs and community-based care models (Chan et al. 2021). Taken together, these pilot cases and the scientific literature on the standardization of longevity clinics constitute a theoretical and comparative framework that any government can use to guide the systematic deployment of public longevity services and to design a national network of public longevity clinics capable of ensuring equitable access.

This is the ideal pathway for public health policy reform: the construction and diffusion of a longevity-based prevention system that anticipates and integrates with the current, traditional healthcare architecture. However, a macro-reform of such magnitude appears difficult to pursue in light of structural and politico-institutional constraints, and potential lack of political will. First, national health systems suffer from path dependency: regulations, clinical routines, and infrastructures developed over decades of reactive medicine complicate reconfiguration toward proactive models (Pierson 2000). Second, the so-called Baumol cost disease means that productivity in the health sector grows more slowly than labor costs, limiting the fiscal space for additional, long-term investment (Baumol & Bowen 1966; OECD 2021). Third, a structural and widespread rollout of longevity clinics would require substantial financial and human resources within a context already marked by shortages of qualified health personnel and mounting demographic pressures (WHO 2016).

At the level of governance, reform would need to coordinate—over an extended period—multiple institutional layers—central government, regions/states, and local authorities—within a protracted process of multi-level policy-making that has historically entailed high transaction costs and political conflict (Kuhlmann & Pavolini 2019; Saltman et al. 2011). In addition, the standardization of new biomarker-based protocols raises issues of ethical regulation, privacy, and data governance that require shared regulatory frameworks and outcome-evaluation

systems still at an embryonic stage (Maier 2024). Social and professional consent must not be overlooked: the introduction of predictive medicine models may encounter cultural resistance among citizens—wary of medicalization—and among clinicians—concerned about potential professional misalignments (Tuohy 1999; Glied & Miller 2015).

Given fiscal constraints, institutional friction, regulatory complexity, and the heavy human-capital requirements, a publicly led, longevity-oriented prevention strategy is more a long-term strategic horizon than a reform that large and mid-sized countries can implement overnight. In this setting, a pragmatic path forward is a diversified policy mix—one that couples innovation with fiscal discipline and expands access—so the longevity megatrend can advance in ways that are both workable and equitable.

3) A Viable Hybrid Policy Approach to Reform Public Health

Building on the challenges outlined in previous sections, the diffusion of the longevity agenda is unlikely to come from a single, capital-intensive macro-policy. In large and administratively complex countries, progress tends to emerge from hybrid policy mixes—coherent bundles of regulatory, fiscal, organizational, and informational instruments. The policy-mix literature demonstrates that complex missions are better served by complementary tools rather than relying on a “hero” instrument (Howlett & Rayner 2007; Rogge & Reichardt 2016; Kern, Rogge & Howlett 2019).

3.1 How Hybrid Mixes Address the Practical Barriers

Multi-level coordination without paralysis. Polycentric governance, as opposed to the multi-level policy making of macro-reforms (Kuhlmann & Pavolini 2019; Saltman et al. 2011), allows different centers to set a clear longevity direction while adhering to common guidelines, standards, and best practices—e.g., community-based screening, integrated care hubs, and mobility/housing adaptations. Polycentric systems, while guided, empirically learn faster and diffuse innovations more reliably than purely centralized rollouts (Ostrom 2010).

Policy layering without path dependency. Institutional reforms are typically incremental, adding new layers rather than replacing existing arrangements. As noted, there is a risk of path dependency, whereby rules, clinical routines, and infrastructures built over decades of reactive medicine complicate structural reconfiguration toward proactive models (Pierson 2000). Structural barriers to rapid, system-wide shifts also include Baumol’s cost disease, where productivity growth in labor-intensive care lags behind rising wages, and fragmented multi-level governance that complicates standardization and scale (Baumol & Bowen 1966). Sequencing interventions—starting with information instruments (nodality) and local pilots before scaling financing—generates learning with limited fiscal exposure while preparing the workforce for broader implementation. Political economy strategies such as stakeholder coalitions, pilot-to-scale roadmaps, and transparent performance metrics are essential to lock in early gains and build public trust around biomarker-driven prevention paradigms.

Cost realism via prevention portfolios. Hybrid policy approaches—sequencing enablers and then scaling sustained prevention portfolios—align with evidence that such portfolios deliver positive return on investment (Masters et al. 2017). This strengthens the fiscal case for progressively shifting expenditure toward preventive care (OECD 2023).

Gradual workforce readiness as a binding constraint. Longevity policies are unlikely to succeed without creating the right interdisciplinary skill mix. Hybrid mixes pair financing and regulatory reforms with targeted workforce strategies, including expanded education capacity, competency frameworks on healthy ageing, retention incentives, and team-based models (WHO 2016; NASEM 2021).

3.2 The Broad Set of Hybrid Policy Instruments for Longevity Policies

To structure the mix, we adopt Hood’s widely used NATO typology—Nodality, Authority, Treasure, Organization—which classifies how governments influence behaviors via information, rules, money, and delivery capacity. In health and social policy, NATO remains a durable map for combining levers across levels of government and sectors (Hood 1983; Hood & Margetts 2007). This framework provides a tractable way to coordinate multi-domain reforms around the broad spectrum of longevity goals (Howlett et al. 2007).

3.2.1 Treasure

A practical option is a low-budget fiscal strategy that reweights public spending toward prevention and functional outcomes while safeguarding discipline and equity. Risk-adjusted capitation for enrolled primary-care panels—augmented with prevention and healthy-ageing indicators (e.g., midlife screening uptake, frailty identification, falls-risk assessment)—creates continuous incentives for proactive management rather than episodic treatment (NASEM 2021; OECD 2023). Public purchasing and targeted subsidies can accelerate the diffusion of validated, low-cost diagnostics—such as biomarker assays and standardized intrinsic-capacity assessments—once analytical and clinical validity thresholds are met, curbing unregulated “biological-age” marketing (Moqri et al. 2023; Perri et al. 2025). Where statutory or social insurance exists, benefit design should minimize out-of-pocket barriers for evidence-based prevention bundles and post-acute rehabilitation. Competitive micro-grants for municipalities—for community testing, mobility, and physical-activity infrastructure—can crowd in local innovation at modest fiscal cost, provided simple outcome reporting is enforced (Howlett & Rayner 2007; Rogge & Reichardt 2016).

3.2.2 Authority

Regulatory tools create the credibility conditions within which financing and organizational reforms can take hold. Accreditation standards for longevity clinics and integrated teams should spell out scope of practice, required competencies, data governance, and learning-and-research cycles; recent blueprints from academic hospitals and geroscience consortia offer reference architectures that can be adapted to publicly funded systems (Bonnes et al. 2024; Mironov et al. 2024). Coverage and advertising rules should tie reimbursement and claims to validated longevity-biomarker endpoints and functional outcomes, both to protect citizens from misleading offers and to align private incentives with public goals (Moqri et al. 2023; Perri et al. 2025). Given the rapid spread of digital tools, regulation of AI-enabled decision support ought to mandate transparency, bias monitoring, and human oversight within clinical governance structures (Hood & Margetts 2007).

3.2.3 Organization

Organizational tools build the capacity to deliver prevention-led, integrated services at scale through a mixed economy of public and contracted providers. Commissioning should favor community-based integrated-care hubs, supervised “medical-gym” pathways for strength and falls prevention, and closed-loop social-prescribing networks. For infrastructure and technology, public–private partnerships (PPPs) are appropriate when risk allocation is explicit, value for money can be demonstrated, and transparency is enforced; absent these conditions, PPPs tend toward rigidity and rent extraction (Hellowell & Vecchi 2016; WHO Europe 2023; Basabih et al. 2022). Framework agreements with qualified private and third-sector longevity providers can speed the rollout of standardized prevention modules under clinical oversight, provided service-level agreements specify data submission and outcome targets. Public procurement and innovation partnerships are especially well suited to low-cost diagnostics and digital registries that require scale to drive down unit costs.

PPPs allow a middle ground between two issues regarding longevity clinics: public systems often lack dedicated longevity services, while many private clinics remain financially fragile and operationally non-standardized. By aligning risk allocation, transparency, and value-for-money tests, PPPs can leverage private capacity while embedding clinical standards within public oversight (WHO Europe 2023; Aging & Disease 2024).

3.2.4 Nodality

Information instruments—Hood’s “nodality”—shape expectations and behavior through measurement, feedback, and public communication. Region- and provider-level dashboards reporting longevity indicators (e.g., preventable DALYs at ages 40–74, composite intrinsic capacity, falls rates, post-acute functional recovery) can support accountability and learning when indicators are valid, salient, and coupled to improvement support; meta-reviews suggest small but meaningful effects of public reporting under these conditions (Cacace et al. 2019; OECD 2023). Within clinical workflows, electronic prompts for mid-life screening, vaccination, medication review, and frailty flags can alter decision framing with low administrative burden (NASEM 2021). Citizen-facing tools—risk calculators and personalized “ageing-well” plans—should be embedded in public campaigns targeting mid-life cohorts and high-risk groups, using plain-language messaging and guided navigation to accredited services.

3.2.5 Workforce

In most longevity reforms, workforce policy is the primary bottleneck and should be treated as a transversal policy lever. Expanding training slots in geriatrics, rehabilitation, and data-literate primary care—paired with return-to-practice incentives—helps correct both the skill mix and regional maldistribution (WHO 2016; NASEM 2021). Competency frameworks for longevity clinics should codify skills in preventive counselling, multimorbidity management, frailty assessment, and functional goal-setting. Team-based payment uplifts can reward clinics that meet defined staffing and training benchmarks (OECD 2023). Because diffusion and democratization of prevention depend on delivery beyond hospital walls, accredited continuing professional development for non-medical settings—pharmacists, exercise professionals, social workers, and community health workers—can transform gyms, pharmacies, and community centers into credible “places of prevention,” provided clear referral pathways and supervision are in place (WHO 2020; NASEM 2021).

3.3 Sequencing and Pilots

Longevity policy is best delivered as a coherent portfolio of complementary instruments rather than a single grand program. The hybrid approach should sequence interventions to produce early wins that can finance later scaling. Immediate (0–2 years) — Nodality / Information: Roll out midlife screening prompts in primary care, regional exposomic mapping pilots, and provider dashboards that expose preventable high-cost utilization. These are low-cost, quick to implement, and provide measurable process signals for learning. Short term (2–5 years) — Authority and Organization pilots: Implement accreditation standards for pilot longevity clinics; run regionally limited payment experiments (risk-adjusted capitation with prevention bonuses); and establish public-procurement standards for validated diagnostics and digital registries. Evaluate as pragmatic pilots with pre-registered outcomes. Medium term (5+ years) — Treasure and scale: Reweight system financing toward prevention portfolios guided by evidence from pilots; employ performance-linked contracting and targeted subsidies to ensure equitable access; and invest in national cohort infrastructure to underpin causal evaluation. This sequencing balances fiscal realism with the imperative to test novel approaches, closing the loop between experimentation and policy adoption while protecting equity.

Clinic design, expected impacts, and evaluation. A public longevity clinic should be designed as a multidisciplinary, outcome-driven node within primary care. Core elements include standardized risk assessment (biomarker panel + functional test battery), a stepped-care prevention pathway (lifestyle, community rehabilitation, pharmacologic interventions when evidence-based), an integrated digital record linking to registries/cohorts, and a built-in evaluation framework with pre-specified functional and economic endpoints. Pilots should incorporate randomized rollout or pragmatic trial elements to allow causal inference about impact, equity, and cost-effectiveness. Evaluation metrics should include changes in composite intrinsic capacity, incidence of first major chronic disease event, preventable hospital admissions, and cost per QALY saved (Olshansky et al. 2007; Maier 2024). Clinics must be publicly accessible or covered by statutory insurance to ensure population-level impact and universality.

Climate adaptation and exposomic planning. Older adults are disproportionately vulnerable to heatwaves, air pollution episodes, and the indirect effects of extreme events. Exposomic planning must therefore include climate-resilience measures targeted at older populations: cooling centers integrated with community health hubs, heatwave early-warning targeted to high-risk cohorts, and urban greening to reduce heat islands and improve air quality. These interventions deliver both immediate protective effects and long-term healthspan benefits by reducing exacerbations of cardiovascular and respiratory disease in vulnerable groups (Nature Communications 2023; major climate-health reviews).

Implementation roadmap — global recommendations. A realistic roadmap organizes actions by feasibility and potential impact over the course of time, in line with the realistic ability of complex healthcare systems to respond to such a challenging change. Year 0–1: Develop national guidance on biomarker evidence tiers and pilot accreditation standards for longevity clinics; fund 3–5 regional exposomic mapping pilots that combine air-quality, social determinant, and primary-care data. Year 1–3: Launch pragmatic trials using validated surrogate endpoints; implement nodality tools (dashboards, screening prompts) and seed a national longitudinal cohort linking exposomic, multi-omic, and clinical data. Year 3–5: Scale successful prevention payment models and public procurement for validated diagnostics; roll out accredited longevity clinic networks in regions with workforce readiness. Ongoing: Monitor equity impact, ensure price-transparency in diagnostics/therapeutics, and sustain multi-stakeholder governance for data and standards.

Final Considerations

To conclude, the evidence underscores longevity medicine as a cornerstone for the sustainable renewal of primary care. The most plausible pathway is not a single, top-down overhaul but a coherent, incremental hybrid of policies that balance early wins, fiscal realism, and workforce readiness. Along this path, proactive prevention—rooted in validated biomarkers, risk stratification, and integrated care—should be coupled with clear regulatory standards, prevention-oriented financing aligned to functional outcomes, distributed delivery capacity (including tightly governed public–private partnerships), and a structured human-capital strategy that formalizes competencies and supports multidisciplinary, data-literate teams (WHO 2016; NASEM 2021).

Whether pursued through macro-reform or a hybrid policy mix, longevity strategy depends on two main foundational preconditions. National recognition of prevention-oriented longevity medicine as a strategic priority, embedding healthy longevity in fiscal, governance, and public-health planning (WHO 2021; NASEM 2022). Development and diffusion of scientific and operational standards for biomarkers of ageing and the service model of public longevity clinics, ensuring analytic validity, clinical validity, clinical utility, and cost-effectiveness (Moqri et al. 2023; Levine et al. 2018; Moqri et al. 2023).

A national commitment to healthy longevity shifts health-system targets beyond disease incidence to intrinsic capacity, functional ability, and healthy life expectancy. This framing legitimizes investment in longitudinal data infrastructures, workforce upskilling, and primary-care redesign—the very elements on which hybrid policy mixes rely (Kickbusch et al. 2016; OECD 2023). It also aligns with the UN Decade of Healthy Ageing (2021–2030), which emphasizes cross-sector action and measurable gains in functional ability across the life course, and is consistent with the geroscience agenda to target ageing mechanisms to reduce multimorbidity and disability (WHO n.d.; Zhavoronkov 2019).

Recognizing longevity medicine as a national strategy also enables governments to establish a coherent framework for biomarkers of ageing. Such a framework distinguishes exploratory research measures from policy-relevant endpoints and sets standards for evidence thresholds—analytical validity, clinical validity, and functional utility—before integrating biomarkers into routine care and reimbursement models (Moqri et al. 2023; Levine et al. 2018; Bonnes et al. 2024).

Regarding longevity clinics, operational heterogeneity poses a risk to credibility and system-level impact. Publicly accessible clinics should be designed as multidisciplinary, outcome-driven nodes embedded in primary care networks, including standardized risk assessments (biomarker panels plus functional tests), stepped-care prevention pathways (lifestyle, rehabilitation, evidence-based pharmacologic interventions), integrated digital records linking to registries and cohorts, and pre-specified functional and economic evaluation endpoints (Olshansky et al. 2007; Maier 2024; Bonnes et al. 2024). Piloting these clinics with pragmatic trials or randomized rollout elements can generate causal evidence on impact, equity, and cost-effectiveness. International exemplars, such as the Sheba Longevity Center in Israel and Alexandra Hospital in Singapore, demonstrate the feasibility of integrating prevention, personalized medicine, and longitudinal evaluation within public systems (Chan et al. 2021; Bonnes et al. 2024).

Finally, a national longevity strategy must consider sequencing, workforce, and climate-related vulnerabilities. Sequenced interventions—beginning with nodality/information instruments, followed by pilot accreditation and organizational experiments, and culminating in financing scale-up—optimize learning while limiting fiscal exposure and preparing the workforce for broader adoption (Howlett & Rayner 2007; Pierson 2000; Baumol & Bowen 1966). Workforce policies should define competency frameworks, expand geriatrics and rehabilitation training, incentivize interdisciplinary team delivery, and embed continuing professional development for allied practitioners (WHO 2016; NASEM,2021). Climate adaptation measures—heatwave early-warning systems, urban greening, and climate-resilient community hubs—are essential to protect older adults, reduce functional decline, and integrate environmental determinants into longevity planning (Nature Communications / major climate-health reviews).

Taken together, the hybrid policy approach, grounded in rigorous biomarker science, operationalized through pilot clinics, and sequenced via coherent policy instruments, provides a realistic and actionable roadmap for public health reform in the era of longevity. By emphasizing functional outcomes, equity, and evidence-based prevention, governments can systematically extend healthspan, reduce chronic disease burden, and sustain the fiscal and operational viability of public healthcare systems.

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